REQUIREMENT ANALYSIS

Functional & Non-functional Requirements

Medical Diagnosis Center in ServiceNow

Functional Requirements

The following requirements describe the functions expected from the proposed Medical Diagnosis Center solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Patient Registration	Create a Patient Record with the required patient information.
FR-2	Diagnostic Test Catalog	Create/select diagnostic tests with name, code, category, description, price, duration and active status.
FR-3	Appointment Booking & Validation	Create an appointment with diagnostic test, date and time; validate mandatory fields, availability, duplicates, overlaps and configured capacity.
FR-4	Approval & Status Management	Approve/reject appointments and manage Requested, Pending Approval, Confirmed, Sample Collected, Processing, Completed and Report Available states.
FR-5	Laboratory Processing	Allow laboratory staff to process approved appointments and maintain patient/test context.
FR-6	Diagnostic Report Management	Create/link reports to appointments and patients and make completed reports available to authorized users.
FR-7	Email Notifications	Send appointment approval and report availability notifications and support configured booking/reminder/cancellation/payment scenarios.
FR-8	Payment Management	Support test-based pricing, payment status and phase-based online payment/invoice/receipt functionality.
FR-9	Cancellation & Refund	Apply a 10% deduction from Total Amount and automatically populate Refund Amount on configured cancellation.
FR-10	Approval / Rejection Details	Record approval note and Approved By on approval; record rejection reason and clear Approved By on rejection.
FR-11	Security & Access	Provide table, field, role and patient-specific ACL access and restrict laboratory report records.
FR-12	Reports & Dashboard	Provide Appointments Reports and Health Reports Dashboard for appointment, status, approval and test activity.

Non-functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Provide a clear, consistent Medical Diagnosis Center workflow through Service Portal pages and widgets.
NFR-2	Security	Use ServiceNow roles and ACLs to protect sensitive patient and report information.
NFR-3	Reliability	Maintain consistent patient, appointment, test, report and payment records.
NFR-4	Performance	Portal pages, validation and record operations should respond efficiently under normal load.
NFR-5	Scalability	Support growth in patients, appointments, tests, staff, locations and reports.
NFR-6	Maintainability	Use maintainable ServiceNow tables, scripts, widgets, Business Rules, notifications and flows.

User Stories and Acceptance Criteria

User Type	Functional Requirement	Story	User Story	Acceptance Criteria	Priority	Release
Patient	Patient Registration	US-1	As a patient, I want to register my details so that my appointment has correct information.	Required details are captured and saved.	High	Release-1
Patient	Diagnostic Test	US-2	As a patient, I want to view diagnostic tests so that I can choose the correct test.	Configured test details are displayed.	High	Release-1
Patient	Appointment Booking	US-3	As a patient, I want to book a test/date/time so that I can reserve a slot.	Valid appointment is created.	High	Release-1
Patient	Booking Validation	US-4	As a patient, I want conflicting bookings to be blocked so that I receive a correct slot.	Mandatory, capacity and conflict rules are enforced.	High	Release-1
Lab Manager	Approval	US-5	As a Lab Manager, I want to approve/reject appointments so that specialized tests are controlled.	Approval status and notes are recorded.	High	Release-2
Technician	Laboratory Processing	US-6	As a laboratory technician, I want to process approved appointments so that reports can be generated.	Appointment progresses through processing states.	High	Release-2
Patient	Reports	US-7	As a patient, I want to access my completed diagnostic report.	Authorized report is available.	High	Release-2
Administrator	Security	US-8	As an administrator, I want ACLs so that sensitive records are protected.	Unauthorized access is denied.	High	Release-3
Administrator	Analytics	US-9	As an administrator, I want reports and dashboards so that I can monitor operations.	Configured metrics are visible.	Medium	Release-3